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JEE MAIN · QUESTION PAPER

JEE Main 2021 (Online) 25th February Evening Shift

JEE Main Question Paper with Answer Key

25 Feb 2021	90	14	English
HELD ON	QUESTIONS	OUT OF SYLLABUS	LANGUAGE

INSIDE THIS PDF

Chemistry	23 questions
Mathematics	25 questions
Physics	28 questions
Out of syllabus	14 questions
Answer key	at the end

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Chemistry 23 questions

Q1. MCQ (Single Correct Answer)

The correct sequence of reagents used in the preparation of 4-bromo-2-nitroethyl benzene from benzene is :

Open solution

(A) $CH_3COCl/AlCl_3, Br_2/AlBr_3, HNO_3/H_2SO_4, Zn/HCl$

(B) $CH_3COCl/AlCl_3, Zn-Hg/HCl, Br_2/AlBr_3, HNO_3/H_2SO_4$

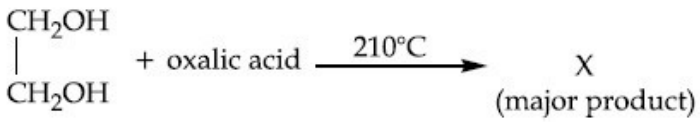
(C) $Br_2/AlBr_3, CH_3COCl/AlCl_3, HNO_3/H_2SO_4, Zn/HCl$

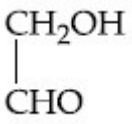
(D) $HNO_3/H_2SO_4, Br_2/AlCl_3, CH_3COCl/AlCl_3, Zn-Hg/HCl$

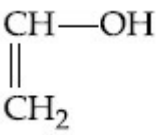
Q2. MCQ (Single Correct Answer)

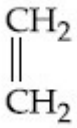
What is 'X' in the given reaction?

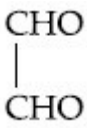
Open solution



(A) 

(B) 

(C) 

(D) 

Q3. MCQ (Single Correct Answer)

The solubility of $Ca(OH)_2$ in water is :

[Given : The solubility product of $Ca(OH)_2$ in water = 5.5×10^{-6}]

Open solution

(A) 1.11×10^{-6} (B) 1.11×10^{-2} (C) 1.77×10^{-6} (D) 1.77×10^{-2}

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Q4. MCQ (Single Correct Answer)

Which among the following species has unequal bond lengths?

(A) XeF_4 (B) BF_4^- (C) SiF_4 (D) SF_4

Open solution

Q5. MCQ (Single Correct Answer)

In which of the following order the given complex ions are arranged correctly with respect to their decreasing spin only magnetic moment?

(i) $[FeF_6]^{3-}$

(ii) $[Co(NH_3)_6]^{3+}$

(iii) $[NiCl_4]^{2-}$

(iv) $[Cu(NH_3)_4]^{2+}$

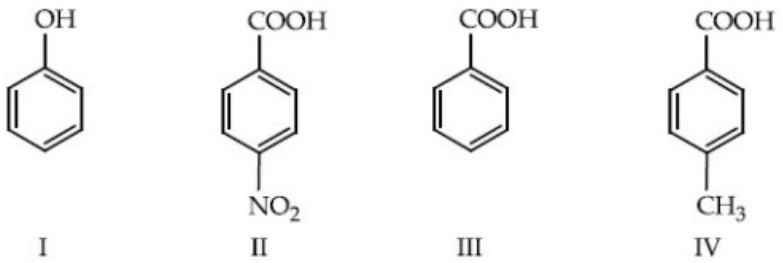
(A) (iii) > (iv) > (ii) > (i) (B) (ii) > (iii) > (i) > (iv)

(C) (i) > (iii) > (iv) > (ii) (D) (ii) > (i) > (iii) > (iv)

Open solution

Q6. MCQ (Single Correct Answer)

The correct order of acid character of the following compounds is :



I II III IV

(A) IV > III > II > I (B) III > II > I > IV

(C) II > III > IV > I (D) I > II > III > IV

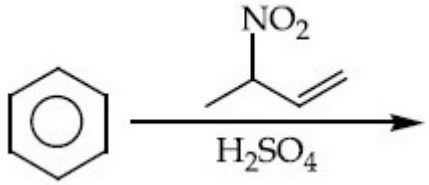
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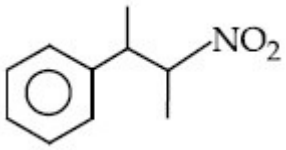
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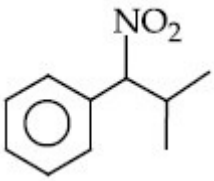
Q7. MCQ (Single Correct Answer)

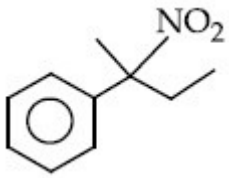
Open solution

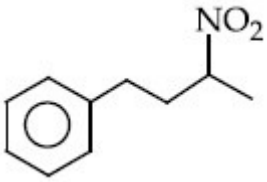
The major product of the following reaction is :



(A) 

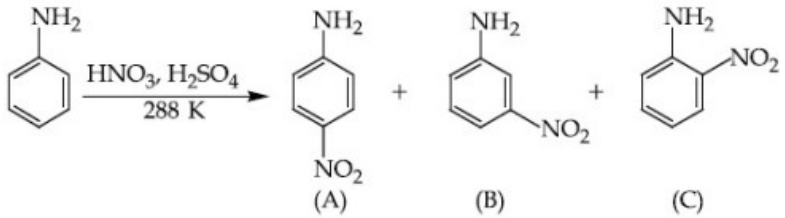
(B) 

(C) 

(D) 

Q8. MCQ (Single Correct Answer)

Open solution



Correct statement about the given chemical reaction is :

(A) $\ddot{N}H_2$ group is *ortho* and *para* directive, so product (B) is not possible.

(B) Reaction is possible and compound (B) will be the major product.

(C) Reaction is possible and compound (A) will be major product.

(D) The reaction will form sulphonated product instead of nitration.

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Q9. MCQ (Single Correct Answer)

Open solution

Given below are two statements :

Statement I : The identification of Ni^{2+} is carried out by dimethyl glyoxime in the presence of NH_4OH.

Statement II : The dimethyl glyoxime is a bidentate neutral ligand.

In the light of the above statements, choose the correct answer from the options given below :

(A) Statement I is false but Statement II is true.

(B) Statement I is true but Statement II is false.

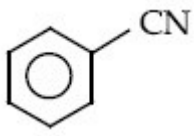
(C) Both Statement I and Statement II are false.

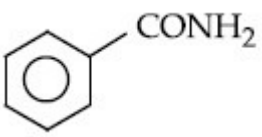
(D) Both Statement I and Statement II are true.

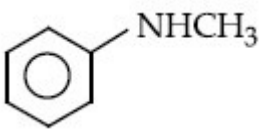
Q10. MCQ (Single Correct Answer)

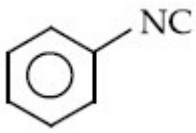
Open solution

Carbylamine test is used to detect the presence of primary amino group in an organic compound. Which of the following compound is formed when this test is performed with aniline?

(A) 

(B) 

(C) 

(D) 

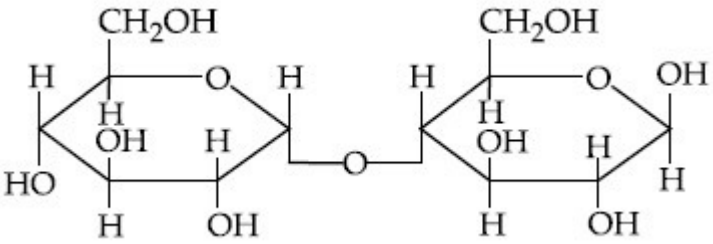
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Q11. MCQ (Single Correct Answer)

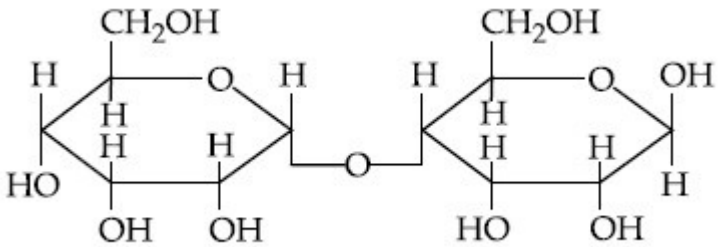
Open solution

Which of the following is correct structure of α-anomer of maltose?

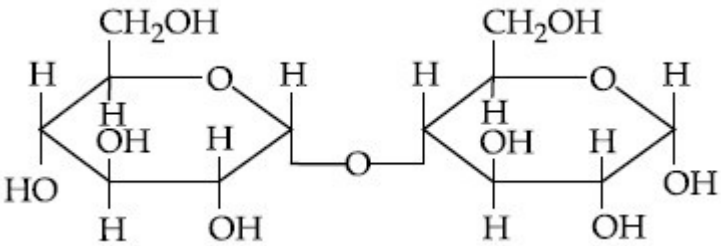
(A)



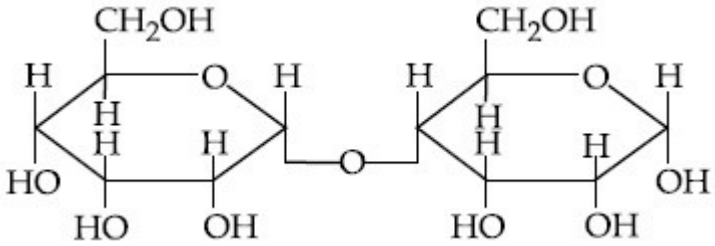
(B)



(C)



(D)



Q12. MCQ (Single Correct Answer)

Open solution

Which of the following compound is added to the sodium extract before addition of silver nitrate for testing of halogens?

(A) Nitric acid

(B) Sodium hydroxide

(C) Ammonia

(D) Hydrochloric acid

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Q13. MCQ (Single Correct Answer)

Open solution

The major product of the following reaction is :

$$CH_3CH_2CH{=}CH_2 \xrightarrow[\text{Rh catalyst}]{H_2/CO}$$

(A) $CH_3CH_2CH_2CHO$

(B) $CH_3CH_2C(=CH_2)CHO$

(C) $CH_3CH_2CH{=}CH{-}CHO$

(D) $CH_3CH_2CH_2CH_2CHO$

Q14. MCQ (Single Correct Answer)

Open solution

The correct order of bond dissociation enthalpy of halogens is :

(A) $F_2 > Cl_2 > Br_2 > I_2$

(B) $Cl_2 > Br_2 > F_2 > I_2$

(C) $I_2 > Br_2 > Cl_2 > F_2$

(D) $Cl_2 > F_2 > Br_2 > I_2$

Q15. Numerical

Open solution

Copper reduces NO_3^- into NO and NO_2 depending upon the concentration of HNO_3 in solution. (Assuming fixed $[Cu^{2+}]$ and $P_{NO} = P_{NO_2}$), the HNO_3 concentration at which the thermodynamic tendency for reduction of NO_3^- into NO and NO_2 by copper is same is 10^x M. The value of 2x is _____. (Rounded off to the nearest integer)

[Given, $E^o_{Cu^{2+}/Cu} = 0.34$ V, $E^o_{NO_3^-/NO} = 0.96$ V, $E^o_{NO_3^-/NO_2} = 0.79$ V and at 298 K, $\frac{RT}{F}(2.303) = 0.059$]

Q16. Numerical

Open solution

The spin only magnetic moment of a divalent ion in aqueous solution (atomic number 29) is _____ BM.

Q17. Numerical

Open solution

If a compound AB dissociates to the extent of 75% in an aqueous solution, the molality of the solution which shows a 2.5 K rise in the boiling point of the solution is _____ molal. (Rounded off to the nearest integer)

[$K_b = 0.52$ K kg mol^{-1}]

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Q18. Numerical

Electromagnetic radiation of wavelength 663 nm is just sufficient to ionise the atom of metal A. The ionization energy of metal A in kJ mol^{-1} is _____. (Rounded off to the nearest integer)

[h = 6.63 $\times$ 10^{-34} Js, c = 3.00 $\times$ 10^{8} ms^{-1}, N_A = 6.02 $\times$ 10^{23} mol^{-1}]

Open solution

Q19. Numerical

Consider titration of NaOH solution versus 1.25 M oxalic acid solution. At the end point following burette readings were obtained.

(i) 4.5 mL

(ii) 4.5 mL

(iii) 4.4 mL

(iv) 4.4 mL

(v) 4.4 mL

If the volume of oxalic acid taken was 10.0 mL then the molarity of the NaOH solution is _____ M. (Rounded off to the nearest integer)

Open solution

Q20. Numerical

Five moles of an ideal gas at 293 K is expanded isothermally from an initial pressure of 2.1 MPa to 1.3 MPa against at constant external pressure 4.3 MPa. The heat transferred in this process is _____ kJ mol^{-1}. (Rounded off to the nearest integer) [Use R = 8.314 J $mol^{-1}K^{-1}$]

Open solution

Q21. Numerical

The number of compound/s given below which contain/s -COOH group is _____. (Integer answer)

(A) Sulphanilic acid

(B) Picric acid

(C) Aspirin

(D) Ascorbic acid

Open solution

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Q22. Numerical

The rate constant of a reaction increases by five times on increase in temperature from 27°C to 52°C. The value of activation energy in kJ mol^{-1} is _____. (Rounded off to the nearest integer)

[R = 8.314 J $K^{-1}mol^{-1}$]

Open solution

Q23. Numerical

Among the following, number of metal/s which can be used as electrodes in the photoelectric cell is _____. (Integer answer)

(A) Li

(B) Na

(C) Rb

(D) Cs

Open solution

Mathematics 25 questions

Q24. MCQ (Single Correct Answer)

The shortest distance between the line x − y = 1 and the curve $x^2 = 2y$ is :

(A) 0 (B) $\frac{1}{2\sqrt{2}}$ (C) $\frac{1}{\sqrt{2}}$ (D) $\frac{1}{2}$

Open solution

Q25. MCQ (Single Correct Answer)

A function f(x) is given by $f(x) = \frac{5^x}{5^x+5}$, then the sum of the series $f\left(\frac{1}{20}\right) + f\left(\frac{2}{20}\right) + f\left(\frac{3}{20}\right) + \ldots\ldots + f\left(\frac{39}{20}\right)$ is equal to :

(A) $\frac{39}{2}$ (B) $\frac{19}{2}$ (C) $\frac{49}{2}$ (D) $\frac{29}{2}$

Open solution

Q26. MCQ (Single Correct Answer)

Let x denote the total number of one-one functions from a set A with 3 elements to a set B with 5 elements and y denote the total number of one-one functions form the set A to the set A × B. Then :

(A) 2y = 273x (B) y = 91x (C) 2y = 91x (D) y = 273x

Open solution

Q27. MCQ (Single Correct Answer)

The integral $\int \frac{e^{3\log_e 2x} + 5e^{2\log_e 2x}}{e^{4\log_e x} + 5e^{3\log_e x} - 7e^{2\log_e x}} dx$, x > 0, is equal to : (where c is a constant of integration)

(A) $\log_e \sqrt{x^2 + 5x - 7} + c$ (B) $4\log_e |x^2 + 5x - 7| + c$

(C) $\frac{1}{4}\log_e |x^2 + 5x - 7| + c$ (D) $\log_e |x^2 + 5x - 7| + c$

Open solution

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Q28. MCQ (Single Correct Answer)

Let A be a set of all 4-digit natural numbers whose exactly one digit is 7. Then the probability that a randomly chosen element of A leaves remainder 2 when divided by 5 is :

(A) $\frac{2}{9}$ (B) $\frac{1}{5}$ (C) $\frac{122}{297}$ (D) $\frac{97}{297}$

Open solution

Q29. MCQ (Single Correct Answer)

Let α and β be the roots of $x^2 - 6x - 2 = 0$. If $a_n = \alpha^n - \beta^n$ for $n \geq 1$, then the value of $\frac{a_{10}-2a_8}{3a_9}$ is :

(A) 3 (B) 2 (C) 4 (D) 1

Open solution

Q30. MCQ (Single Correct Answer)

If $0 < x, y < \pi$ and $\cos x + \cos y - \cos(x + y) = \frac{3}{2}$, then $\sin x + \cos y$ is equal to :

(A) $\frac{1+\sqrt{3}}{2}$ (B) $\frac{1}{2}$ (C) $\frac{\sqrt{3}}{2}$ (D) $\frac{1-\sqrt{3}}{2}$

Open solution

Q31. MCQ (Single Correct Answer)

$\operatorname{cosec}\left[2\cot^{-1}(5) + \cos^{-1}\left(\frac{4}{5}\right)\right]$ is equal to :

(A) $\frac{75}{56}$ (B) $\frac{65}{56}$ (C) $\frac{56}{33}$ (D) $\frac{65}{33}$

Open solution

Q32. MCQ (Single Correct Answer)

Let A be a 3×3 matrix with det(A) = 4. Let R_i denote the ith row of A. If a matrix B is obtained by performing the operation $R_2 \to 2R_2 + 5R_3$ on 2A, then det(B) is equal to :

(A) 64 (B) 16 (C) 128 (D) 80

Open solution

Q33. MCQ (Single Correct Answer)

If $\alpha, \beta \in R$ are such that $1 - 2i$ (here $i^2 = -1$) is a root of $z^2 + \alpha z + \beta = 0$, then $(\alpha - \beta)$ is equal to :

(A) -7 (B) 7 (C) 3 (D) -3

Open solution

Q34. MCQ (Single Correct Answer)

The minimum value of $f(x) = a^{a^x} + a^{1-a^x}$, where a, $x \in R$ and a > 0, is equal to :

(A) $a + \frac{1}{a}$ (B) 2a (C) a + 1 (D) $2\sqrt{a}$

Open solution

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Q35. MCQ (Single Correct Answer)

If $I_n = \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \cot^n x \, dx$, then :

Open solution

(A) $\frac{1}{I_2+I_4}, \frac{1}{I_3+I_5}, \frac{1}{I_4+I_6}$ are in A.P.

(B) $I_2 + I_4, I_3 + I_5, I_4 + I_6$ are in A.P.

(C) $\frac{1}{I_2+I_4}, \frac{1}{I_3+I_5}, \frac{1}{I_4+I_6}$ are in G.P.

(D) $I_2 + I_4, (I_3 + I_5)^2, I_4 + I_6$ are in G.P.

Q36. MCQ (Single Correct Answer)

In a group of 400 people, 160 are smokers and non-vegetarian; 100 are smokers and vegetarian and the remaining 140 are non-smokers and vegetarian. Their chances of getting a particular chest disorder are 35%, 20% and 10% respectively. A person is chosen from the group at random and is found to be suffering from the chest disorder. The probability that the selected person is a smoker and non-vegetarian is :

Open solution

(A) $\frac{14}{45}$ (B) $\frac{8}{45}$ (C) $\frac{7}{45}$ (D) $\frac{28}{45}$

Q37. MCQ (Single Correct Answer)

If for the matrix, $A = \begin{bmatrix} 1 & -\alpha \\ \alpha & \beta \end{bmatrix}$, $AA^T = I_2$, then the value of $\alpha^4 + \beta^4$ is :

Open solution

(A) 3 (B) 2 (C) 1 (D) 4

Q38. MCQ (Single Correct Answer)

If the curve $x^2 + 2y^2 = 2$ intersects the line $x + y = 1$ at two points P and Q, then the angle subtended by the line segment PQ at the origin is :

Open solution

(A) $\frac{\pi}{2} - \tan^{-1}\left(\frac{1}{4}\right)$ (B) $\frac{\pi}{2} + \tan^{-1}\left(\frac{1}{3}\right)$

(C) $\frac{\pi}{2} - \tan^{-1}\left(\frac{1}{3}\right)$ (D) $\frac{\pi}{2} + \tan^{-1}\left(\frac{1}{4}\right)$

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Q39. MCQ (Single Correct Answer)

The following system of linear equations

$2x + 3y + 2z = 9$

$3x + 2y + 2z = 9$

$x - y + 4z = 8$

(A) does not have any solution

(B) has a solution (α, β, γ) satisfying $\alpha + \beta^2 + \gamma^3 = 12$

(C) has a unique solution

(D) has infinitely many solutions

Open solution

Q40. MCQ (Single Correct Answer)

A hyperbola passes through the foci of the ellipse $\frac{x^2}{25} + \frac{y^2}{16} = 1$ and its transverse and conjugate axes coincide with major and minor axes of the ellipse, respectively. If the product of their eccentricities is one, then the equation of the hyperbola is :

(A) $\frac{x^2}{9} - \frac{y^2}{4} = 1$

(B) $\frac{x^2}{9} - \frac{y^2}{16} = 1$

(C) $\frac{x^2}{9} - \frac{y^2}{25} = 1$

(D) $x^2 - y^2 = 9$

Open solution

Q41. Numerical

If the remainder when x is divided by 4 is 3, then the remainder when $(2020 + x)^{2022}$ is divided by 8 is _____.

Open solution

Q42. Numerical

The total number of two digit numbers 'n', such that $3^n + 7^n$ is a multiple of 10, is _____.

Open solution

Q43. Numerical

The value of $\int_{-2}^{2} |3x^2 - 3x - 6| dx$ is _____.

Open solution

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Q44. Numerical

A line 'l' passing through origin is perpendicular to the lines

$l_1 : \vec{r} = (3+t)\hat{i} + (-1+2t)\hat{j} + (4+2t)\hat{k}$

$l_2 : \vec{r} = (3+2s)\hat{i} + (3+2s)\hat{j} + (2+s)\hat{k}$

If the co-ordinates of the point in the first octant on 'l_2' at a distance of $\sqrt{17}$ from the point of intersection of 'l' and 'l_1' are (a, b, c) then 18(a + b + c) is equal to _____.

Open solution

Q45. Numerical

A function f is defined on $[-3, 3]$ as

$f(x) = \begin{cases} \min\{|x|, 2-x^2\}, & -2 \le x \le 2 \\ [|x|], & 2 < |x| \le 3 \end{cases}$ where [x] denotes the greatest integer $\le$ x. The number of points, where f is not differentiable in $(-3, 3)$ is _____.

Open solution

Q46. Numerical

Let $\vec{a} = \hat{i} + \alpha\hat{j} + 3\hat{k}$ and $\vec{b} = 3\hat{i} - \alpha\hat{j} + \hat{k}$. If the area of the parallelogram whose adjacent sides are represented by the vectors $\vec{a}$ and $\vec{b}$ is $8\sqrt{3}$ square units, then $\vec{a} \cdot \vec{b}$ is equal to _____.

Open solution

Q47. Numerical

If $\lim\limits_{x\to 0} \frac{ax-(e^{4x}-1)}{ax(e^{4x}-1)}$ exists and is equal to b, then the value of a $-$ 2b is _____.

Open solution

Q48. Numerical

If the curve, y = y(x) represented by the solution of the differential equation $(2xy^2 - y)dx + xdy = 0$, passes through the intersection of the lines, $2x - 3y = 1$ and $3x + 2y = 8$, then $|y(1)|$ is equal to _____.

Open solution

Physics 28 questions

Q49. MCQ (Single Correct Answer)

An LCR circuit contains resistance of 110Ω and a supply of 220 V at 300 rad/s angular frequency. If only capacitance is removed from the circuit, current lags behind the voltage by 45°. If on the other hand, only inductor is removed the current leads by 45° with the applied voltage. The rms current flowing in the circuit will be :

(A) 1 A (B) 2.5 A (C) 2 A (D) 1.5 A

Open solution

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Q50. MCQ (Single Correct Answer)

An electron with kinetic energy K_1 enters between parallel plates of a capacitor at an angle 'α' with the plates. It leaves the plates at angle 'β' with kinetic energy K_2. Then the ratio of kinetic energies $K_1 : K_2$ will be :

Open solution

(A) $\frac{\cos^2\beta}{\cos^2\alpha}$

(B) $\frac{\cos\beta}{\cos\alpha}$

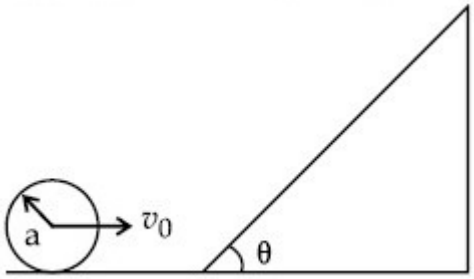
(C) $\frac{\sin^2\beta}{\cos^2\alpha}$

(D) $\frac{\cos\beta}{\sin\alpha}$

Q51. MCQ (Single Correct Answer)

A sphere of radius 'a' and mass 'm' rolls along a horizontal plane with constant speed v_0. It encounters an inclined plane at angle θ and climbs upward. Assuming that it rolls without slipping, how far up the sphere will travel?

Open solution



(A) $\frac{v_0^2}{2g\sin\theta}$

(B) $\frac{7v_0^2}{10g\sin\theta}$

(C) $\frac{2}{5}\frac{v_0^2}{g\sin\theta}$

(D) $\frac{v_0^2}{5g\sin\theta}$

Q52. MCQ (Single Correct Answer)

An electron of mass m_e and a proton of mass m_p = 1836 m_e are moving with the same speed. The ratio of their de Broglie wavelength $\frac{\lambda_{electron}}{\lambda_{proton}}$ will be :

Open solution

(A) 1

(B) 1836

(C) $\frac{1}{1836}$

(D) 918

Q53. MCQ (Single Correct Answer)

Consider the diffraction pattern obtained from the sunlight incident on a pinhole of diameter 0.1 μm. If the diameter of the pinhole is slightly increased, it will affect the diffraction pattern such that:

Open solution

(A) its size increases, but intensity decreases

(B) its size increases, and intensity increases

(C) its size decreases, but intensity increases

(D) its sizes decreases, and intensity decreases

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Q54. MCQ (Single Correct Answer)

Open solution

In a ferromagnetic material, below the curie temperature, a domain is defined as :

(A) a macroscopic region with zero magnetization.

(B) a macroscopic region with saturation magnetization.

(C) a macroscopic region with randomly oriented magnetic dipoles.

(D) a macroscopic region with consecutive magnetic dipoles oriented in opposite direction.

Q55. MCQ (Single Correct Answer)

Open solution

Given below are two statements :

Statement I : In a diatomic molecule, the rotational energy at a given temperature obeys Maxwell's distribution.

Statement II : In a diatomic molecule, the rotational energy at a given temperature equals the translational kinetic energy for each molecule.

In the light of the above statements, choose the correct answer from the options given below :

(A) Both Statement I and Statement II are true

(B) Both Statement I and Statement II are false

(C) Statement I is true but Statement II is false.

(D) Statement I is false but Statement II is true.

Q56. MCQ (Single Correct Answer)

Open solution

Match List I with List II.

	List I		List II
(a)	Rectifier	(i)	Used either for stepping up or stepping down the a.c. voltage
(b)	Stabilizer	(ii)	Used to convert a.c. voltage into d.c. voltage
(c)	Transformer	(iii)	Used to remove any ripple in the rectified output voltage
(d)	Filter	(iv)	Used for constant output voltage even when the input voltage or load current change

Choose the correct answer from the options given below :

(A) (a)-(ii), (b)-(iv), (c)-(i), (d)-(iii)

(B) (a)-(iii), (b)-(iv), (c)-(i), (d)-(ii)

(C) (a)-(ii), (b)-(i), (c)-(iv), (d)-(iii)

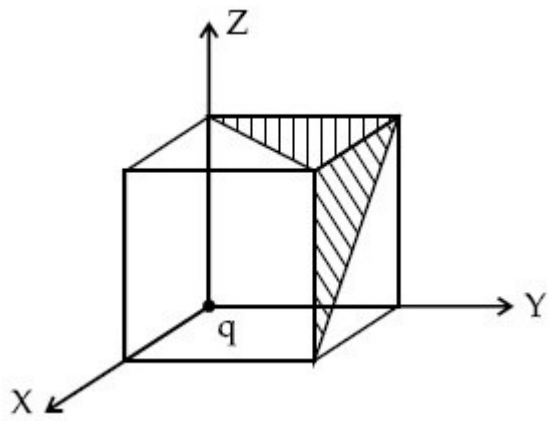
(D) (a)-(ii), (b)-(i), (c)-(iii), (d)-(iv)

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Q57. MCQ (Single Correct Answer)

A charge 'q' is placed at one corner of a cube as shown in figure. The flux of electrostatic field $\vec{E}$ through the shaded area is :

Open solution

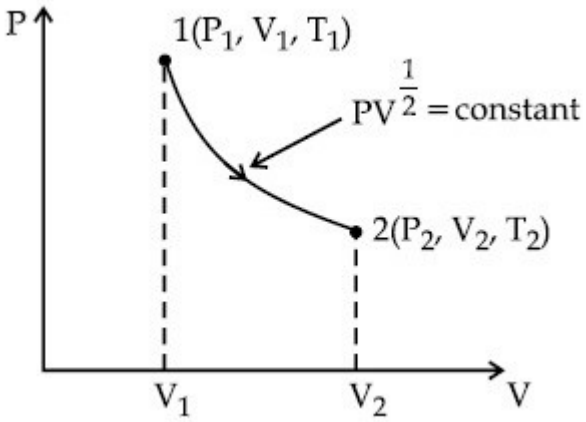


(A) $\frac{q}{24\varepsilon_0}$ (B) $\frac{q}{48\varepsilon_0}$ (C) $\frac{q}{4\varepsilon_0}$ (D) $\frac{q}{8\varepsilon_0}$

Q58. MCQ (Single Correct Answer)

Thermodynamic process is shown below on a P-V diagram for one mole of an ideal gas. If $V_2 = 2V_1$ then the ratio of temperature T_2/T_1 is :

Open solution



(A) $\sqrt{2}$ (B) $\frac{1}{\sqrt{2}}$ (C) $\frac{1}{2}$ (D) 2

Q59. MCQ (Single Correct Answer)

The wavelength of the photon emitted by a hydrogen atom when an electron makes a transition from n = 2 to n = 1 state is :

Open solution

(A) 194.8 nm (B) 490.7 nm (C) 913.3 nm (D) 121.8 nm

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Q60. MCQ (Single Correct Answer)

Y = A sin(ωt + ϕ_0) is the time-displacement equation of a SHM. At t = 0 the displacement of the particle is $Y = \frac{A}{2}$ and it is moving along negative x-direction. Then the initial phase angle ϕ_0 will be:

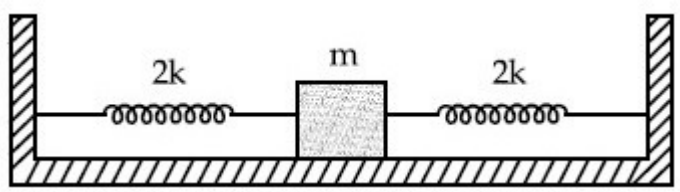
Open solution

(A) $\frac{5\pi}{6}$ (B) $\frac{\pi}{3}$ (C) $\frac{2\pi}{3}$ (D) $\frac{\pi}{6}$

Q61. MCQ (Single Correct Answer)

Two identical springs of spring constant '2k' are attached to a block of mass m and to fixed support (see figure). When the mass is displaced from equilibrium position on either side, it executes simple harmonic motion. The time period of oscillations of this system is :

Open solution

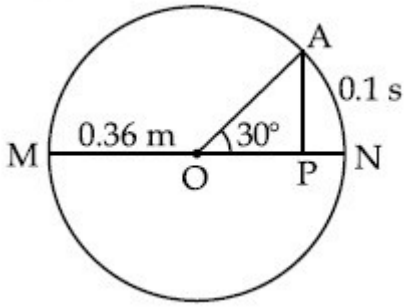


(A) $2\pi\sqrt{\frac{m}{k}}$ (B) $\pi\sqrt{\frac{m}{k}}$ (C) $2\pi\sqrt{\frac{m}{2k}}$ (D) $\pi\sqrt{\frac{m}{2k}}$

Q62. MCQ (Single Correct Answer)

The point A moves with a uniform speed along the circumference of a circle of radius 0.36 m and covers 30° in 0.1 s. The perpendicular projection 'P' from 'A' on the diameter MN represents the simple harmonic motion of 'P'. The restoration force per unit mass when P touches M will be :

Open solution



(A) 9.87 N (B) 0.49 N (C) 50 N (D) 100 N

Q63. MCQ (Single Correct Answer)

If e is the electronic charge, c is the speed of light in free space and h is Planck's constant, the quantity $\frac{1}{4\pi\varepsilon_0}\frac{|e|^2}{hc}$ has dimensions of :

Open solution

(A) $[MLT^{-1}]$ (B) $[MLT^0]$ (C) $[M^0L^0T^0]$ (D) $[LC^{-1}]$

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Q64. MCQ (Single Correct Answer)

Open solution

For extrinsic semiconductors; when doping level is increased;

(A) Fermi-level of both p-type and n-type semiconductors will go upward for $T > T_F K$ and downward for $T < T_F K$, where T_F is Fermi temperature.

(B) Fermi-level of p-type semiconductor will go upward and Fermi-level of n-type semiconductors will go downward

(C) Fermi-level of p and n-type semiconductors will not be affected.

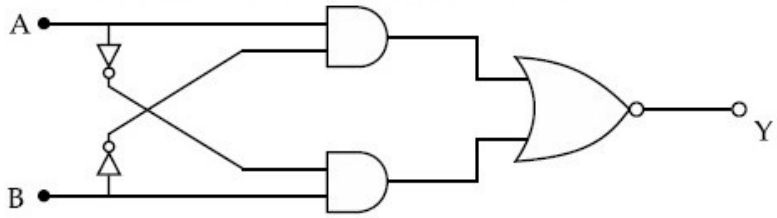
(D) Fermi-level of p-type semiconductors will go downward and Fermi-level of n-type semiconductor will go upward.

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Q65. MCQ (Single Correct Answer)

Open solution

The truth table for the following logic circuit is :



(A)

A	B	Y
0	0	0
0	1	1
1	0	1
1	1	0

(B)

A	B	Y
0	0	1
0	1	0
1	0	1
1	1	0

(C)

A	B	Y
0	0	0
0	1	1
1	0	0
1	1	1

(D)

A	B	Y
0	0	1
0	1	0
1	0	0
1	1	1

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Q66. MCQ (Single Correct Answer)

A stone is dropped from the top of a building. When it crosses a point 5 m below the top, another stone starts to fall from a point 25 m below the top. Both stones reach the bottom of building simultaneously. The height of the building is :

(A) 50 m (B) 25 m (C) 45 m (D) 35 m

Open solution

Q67. MCQ (Single Correct Answer)

The stopping potential for electrons emitted from a photosensitive surface illuminated by light of wavelength 491 nm is 0.710 V. When the incident wavelength is changed to a new value, the stopping potential is 1.43 V. The new wavelength is :

(A) 400 nm (B) 329 nm (C) 309 nm (D) 382 nm

Open solution

Q68. Numerical

If $\vec{P} \times \vec{Q} = \vec{Q} \times \vec{P}$, the angle between $\vec{P}$ and $\vec{Q}$ is $\theta(0^\circ < \theta < 360^\circ)$. The value of '$\theta$' will be _____$^\circ$.

Open solution

Q69. Numerical

The percentage increase in the speed of transverse waves produced in a stretched string if the tension is increased by 4%, will be _____%.

Open solution

Q70. Numerical

The initial velocity v_i required to project a body vertically upward from the surface of the earth to reach a height of 10R, where R is the radius of the earth, may be described in terms of escape velocity v_e such that $v_i = \sqrt{\frac{x}{y}} \times v_e$. The value of x will be _____.

Open solution

Q71. Numerical

The peak electric field produced by the radiation coming from the 8W bulb at a distance of 10 m is $\frac{x}{10}\sqrt{\frac{\mu_0 c}{\pi}}\frac{V}{m}$. The efficiency of the bulb is 10% and it is a point source. The value of x is _____.

Open solution

Q72. Numerical

Two identical conducting spheres with negligible volume have 2.1 nC and -0.1 nC charges, respectively. They are brought into contact and then separated by a distance of 0.5 m. The electrostatic force acting between the spheres is _____ $\times 10^{-9}$ N.

[Given : $4\pi\varepsilon_0 = \frac{1}{9\times10^9}$ SI unit]

Open solution

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Q73. Numerical

The wavelength of an X-ray beam is $10\overset{o}{A}$. The mass of a fictitious particle having the same energy as that of the X-ray photons is $\frac{x}{3}h$ kg. The value of x is _____. (h = Planck's constant)

Open solution

Q74. Numerical

Two small spheres each of mass 10 mg are suspended from a point by threads 0.5 m long. They are equally charged and repel each other to a distance of 0.20 m. The charge on each of the sphere is $\frac{a}{21} \times 10^{-8}$C. The value of 'a' will be _____. [Given g = 10 ms^{-2}]

Open solution

Q75. Numerical

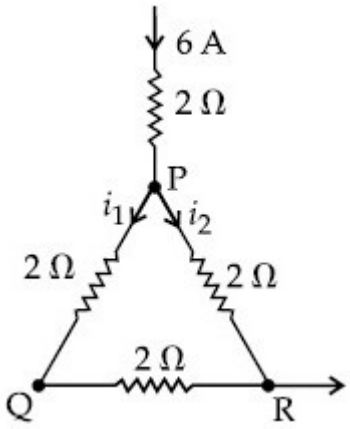
Two particles having masses 4 g and 16 g respectively are moving with equal kinetic energies. The ratio of the magnitudes of their linear momentum is n : 2. The value of n will be _____.

Open solution

Q76. Numerical

A current of 6A enters one corner P of an equilateral triangle PQR having 3 wires of resistance 2Ω each and leaves by the corner R. The currents i_1 in ampere is _____.

Open solution



Out of syllabus 14 questions

These questions are on topics no longer in the current syllabus. They are kept for practice, after everything else.

Q77. MCQ (Single Correct Answer) · **Chemistry**

Water does not produce CO on reacting with :

(A) C (B) CH_4 (C) C_3H_8 (D) CO_2

Open solution

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Q78. MCQ (Single Correct Answer) · **Chemistry**

Open solution

Given below are two statements :

Statement I : The pH of rain water is normally 5.6.

Statement II : If the pH of rain water drops below 5.6, it is called acid rain.

In the light of the above statements, choose the correct answer from the options given below :

(A) Both Statement I and Statement II are true.

(B) Both Statement I and Statement II are false.

(C) Statement I is true but Statement II is false.

(D) Statement I is false but Statement II is true.

Q79. MCQ (Single Correct Answer) · **Chemistry**

Open solution

Given below are two statements :

Statement I : α and β forms of sulphur can change reversibly between themselves with slow heating or slow cooling.

Statement II : At room temperature the stable crystalline form of sulphur is monoclinic sulphur.

In the light of the above statements, choose the correct answer from the options given below :

(A) Statement I is true but Statement II is false.

(B) Statement I is false but Statement II is true.

(C) Both Statement I and Statement II are true.

(D) Both Statement I and Statement II are false.

Q80. MCQ (Single Correct Answer) · **Chemistry**

Open solution

The major components of German Silver are :

(A) Zn, Ni and Ag (B) Cu, Zn and Ni (C) Cu, Zn and Ag (D) Ge, Cu and Ag

Q81. MCQ (Single Correct Answer) · **Chemistry**

Open solution

The method used for the purification of Indium is :

(A) zone refining (B) vapour phase refining

(C) liquation (D) van Arkel method

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Q82. MCQ (Single Correct Answer) · **Chemistry**

Which one of the following statements is FALSE for hydrophilic sols?

(A) They do not require electrolytes for stability.

(B) The sols cannot be easily coagulated.

(C) These sols are reversible in nature.

(D) Their viscosity is of the order of that of H_2O.

Open solution

Q83. Numerical · **Chemistry**

The unit cell of copper corresponds to a face centered cube of edge length 3.596 $\overset{o}{A}$ with one copper atom at each lattice point. The calculated density of copper in kg/m^3 is _____. [Molar mass of Cu : 63.54 g; Avogadro Number = 6.022×10^{23}]

Open solution

Q84. MCQ (Single Correct Answer) · **Mathematics**

The contrapositive of the statement "If you will work, you will earn money" is :

(A) If you will not earn money, you will not work

(B) If you will earn money, you will work

(C) You will earn money, if you will not work

(D) To earn money, you need to work

Open solution

Q85. MCQ (Single Correct Answer) · **Mathematics**

A plane passes through the points A(1, 2, 3), B(2, 3, 1) and C(2, 4, 2). If O is the origin and P is (2, -1, 1), then the projection of $\overrightarrow{OP}$ on this plane is of length :

(A) $\sqrt{\frac{2}{7}}$ (B) $\sqrt{\frac{2}{5}}$ (C) $\sqrt{\frac{2}{3}}$ (D) $\sqrt{\frac{2}{11}}$

Open solution

Q86. MCQ (Single Correct Answer) · **Mathematics**

$\lim\limits_{n\to\infty}\left[\frac{1}{n}+\frac{n}{(n+1)^2}+\frac{n}{(n+2)^2}+\ldots\ldots\ldots+\frac{n}{(2n-1)^2}\right]$ is equal to :

(A) $\frac{1}{2}$ (B) $\frac{1}{3}$ (C) 1 (D) $\frac{1}{4}$

Open solution

Q87. Numerical · **Mathematics**

A line is a common tangent to the circle $(x-3)^2 + y^2 = 9$ and the parabola $y^2 = 4x$. If the two points of contact (a, b) and (c, d) are distinct and lie in the first quadrant, then 2(a + c) is equal to _____.

Open solution

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Q88. Numerical · **Mathematics**

If the curves $x = y^4$ and $xy = k$ cut at right angles, then $(4k)^6$ is equal to _____.

Open solution

Q89. MCQ (Single Correct Answer) · **Physics**

If a message signal of frequency 'f_m' is amplitude modulated with a carrier signal of frequency 'f_c' and radiated through an antenna, the wavelength of the corresponding signal in air is :

(A) $\frac{c}{f_c+f_m}$ (B) $\frac{c}{f_c}$ (C) $\frac{c}{f_c-f_m}$ (D) $\frac{c}{f_m}$

Open solution

Q90. Numerical · **Physics**

A reversible heat engine converts one-fourth of the heat input into work. When the temperature of the sink is reduced by 52K, its efficiency is doubled. The temperature in Kelvin of the source will be _____.

Open solution

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Answer Key

Chemistry

1. **B**	2. **C**	3. **B**	4. **D**	5. **C**
6. **C**	7. **A**	8. **C**	9. **D**	10. **D**
11. **C**	12. **A**	13. **D**	14. **B**	15. **1**
16. **2**	17. **3**	18. **181**	19. **6**	20. **15**
21. **1**	22. **52**	23. **1**		

Mathematics

24. **B**	25. **A**	26. **C**	27. **B**	28. **D**
29. **B**	30. **A**	31. **B**	32. **A**	33. **A**
34. **D**	35. **A**	36. **D**	37. **C**	38. **D**
39. **C**	40. **B**	41. **1**	42. **45**	43. **19**
44. **44**	45. **5**	46. **2**	47. **5**	48. **1**

Physics

49. **C**	50. **A**	51. **B**	52. **B**	53. **C**
54. **B**	55. **C**	56. **A**	57. **A**	58. **A**
59. **D**	60. **A**	61. **B**	62. **A**	63. **C**
64. **D**	65. **D**	66. **C**	67. **D**	68. **180**
69. **2**	70. **10**	71. **2**	72. **36**	73. **10**
74. **20**	75. **1**	76. **2**		

Out of syllabus

77. **D**	78. **A**	79. **A**	80. **B**	81. **A**
82. **D**	83. **9077**	84. **A**	85. **D**	86. **A**
87. **9**	88. **4**	89. **B**	90. **208**	

For numerical answers given as a range, any value within the range is correct. Full solutions: scan the QR code next to each question, or visit questions.examside.com.

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